

Here are the notes from the video, presented in a more formal style suitable for technical documentation, including LaTeX for mathematical expressions.

Subject: Linear Regression (General Linear Models)

Source: StatQuest with Josh Starmer <http://www.youtube.com/watch?v=7ArmBVF2dCs>

I. Core Concepts of Linear Regression

Linear Regression analysis is primarily concerned with three main objectives:

1. **Model Fitting:** Employing the method of **Least Squares** to fit a line (or hyperplane) to the dataset.
2. **Goodness-of-Fit:** Calculating R^2 (**R-squared**), the coefficient of determination, to quantify the proportion of variance explained by the model.
3. **Statistical Significance:** Determining the **p-value** associated with R^2 to assess the reliability of the model and whether the observed relationship is likely due to chance.

II. Part 1: The Method of Least Squares

The objective of the least squares method is to find the optimal parameters (intercept and slope) for a line that best fits the data. This "best fit" is defined as the line that minimizes the total squared error.

- **Dataset Example:** Predicting mouse **size** (dependent variable, Y) using mouse **weight** (independent variable, X).
- **Residual:** The vertical distance between an observed data point (y_i) and the point on the fitted line ($\hat{y}_i$). This represents the prediction error for that data point.
 - $\text{Residual}_i = y_i - \hat{y}_i$
- **Least Squares Process [00:01:50 - 00:02:56]:**
 - i. A line is posited through the data.
 - ii. The residual for each data point is calculated.
 - iii. Each residual is squared.
 - iv. All squared residuals are summed to get the **Sum of Squared Residuals (SSR)**, also known as the Sum of Squares around the Fit (SS_{fit}).
 - $SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
 - v. The parameters of the line (slope and intercept) are iteratively adjusted (e.g., "rotated") to find the specific parameters that result in the minimum possible SSR .
- **Resulting Equation:** The method estimates the parameters β_0 (y-intercept) and β_1 (slope) for the linear equation:

$$Y = \beta_0 + \beta_1 X$$

A non-zero slope ($\beta_1 \neq 0$) implies that X provides predictive information for Y .

III. Part 2: R-squared (R^2), The Coefficient of Determination

R^2 measures how well the independent variable(s) explain the variation in the dependent variable. It is a value between 0 and 1.

- **Step 1: Calculate Total Variation (Baseline)**
 - Calculate the mean of the dependent variable (e.g., average mouse size, $\bar{y}$).
 - Measure the squared distance from each data point (y_i) to this mean.
 - The sum of these squared distances is the **Total Sum of Squares (SS_{total})**, also referred to in the video as SS_{mean} . This represents the total variation in Y *without* considering X .
 - $SS_{\text{total}} = \sum_{i=1}^n (y_i - \bar{y})^2$

- **Step 2: Calculate Unexplained Variation (Model Error)**

- This is the Sum of Squared Residuals (SSR or SS_{fit}) from the least squares fit, as calculated in Part 1. It represents the variation that *remains unexplained* by the model.

$$SS_{\text{fit}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **Step 3: Calculate R^2**

- R^2 is the proportion of the total variation that is *explained* by the model. It is calculated by finding the reduction in variance from the mean-only model to the fitted model, and expressing it as a fraction of the total variance.

- **Formula:**

$$R^2 = \frac{SS_{\text{total}} - SS_{\text{fit}}}{SS_{\text{total}}} = 1 - \frac{SS_{\text{fit}}}{SS_{\text{total}}}$$

- **Interpreting R^2 Values:**

- $R^2 = 1$: Perfect fit. The model explains 100% of the variation in Y . All data points lie on the fitted line, and $SS_{\text{fit}} = 0$.
- $R^2 = 0$: Useless fit. The model explains 0% of the variation in Y . The fitted line is simply the horizontal mean line ($\beta_1 = 0$), and $SS_{\text{fit}} = SS_{\text{total}}$.
- $R^2 = 0.6$: The model explains 60% of the variation in Y .

- **Multiple Regression & Adjusted R^2**

- The R^2 concept applies to models with multiple predictors (e.g., $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$).
- **Problem:** Adding *any* new predictor, even a random one (like "astrological sign"), will *never* decrease R^2 and will almost always increase it slightly due to random chance. This can lead to overfitting.
- **Solution:** **Adjusted R^2** is used. This metric penalizes the R^2 value for each additional parameter (k) added to the model, relative to the sample size (n). It provides a more honest assessment of model fit when comparing models with different numbers of predictors.

$$R_{\text{adj}}^2 = 1 - \left(\frac{(1-R^2)(n-1)}{n-k-1} \right)$$

IV. Part 3: Statistical Significance (P-value)

The p-value determines if the calculated R^2 value is statistically significant, or if the observed relationship could have arisen purely by random chance.

- **The Problem with R^2 Alone:** With only two data points, R^2 will *always* be 1, but this result is meaningless.
- **The F-statistic:** The p-value for the overall model's R^2 is derived from an **F-statistic**, which is a ratio of explained to unexplained variance, adjusted for degrees of freedom.
- **Conceptual Formula:**

$$F = \frac{\text{Variation explained by the model}}{\text{Variation not explained by the model}}$$

- **Formal Calculation:**

The F-statistic is the ratio of the **Mean Square Regression (MSR)** to the **Mean Square Error (MSE)**.

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{(SS_{\text{total}} - SS_{\text{fit}})/(p_{\text{fit}} - p_{\text{mean}})}{SS_{\text{fit}}/(n - p_{\text{fit}})}$$

- **Where:**

- n = sample size.
- SS_{total} = Total Sum of Squares (variation around the mean).
- SS_{fit} = Sum of Squares Residual (unexplained variation).
- p_{fit} = number of parameters in the fitted model (e.g., 2 for $Y = \beta_0 + \beta_1 X$).
- p_{mean} = number of parameters in the null (mean-only) model (1, for β_0).

- **Numerator:** The explained variance, divided by its degrees of freedom ($p_{\text{fit}} - p_{\text{mean}}$), which is the number of new parameters added.
- **Denominator:** The unexplained variance (residuals), divided by its degrees of freedom ($n - p_{\text{fit}}$).

- **Deriving the P-value:**

- A high F-statistic implies the model explains a large amount of variance relative to its unexplained variance.
- This F-statistic is compared to an **F-distribution**, which represents the distribution of F-statistics that would occur by random chance if the null hypothesis ($H_0 : \beta_1 = 0$) were true.

iii. The **p-value** is the probability of observing an F-statistic as large as or larger than the one calculated, given that the null hypothesis is true.

V. Conclusion

A robust linear regression result requires two conditions to be met:

1. **A large R^2 value:** Indicates that the model *quantifies* a substantial portion of the variation in the dependent variable.
2. **A small p-value:** Indicates that the relationship is *reliable* and unlikely to be a product of random sampling chance.